114 Fall Semester **Special Topics in RL, NTUST ME** **Final Project**

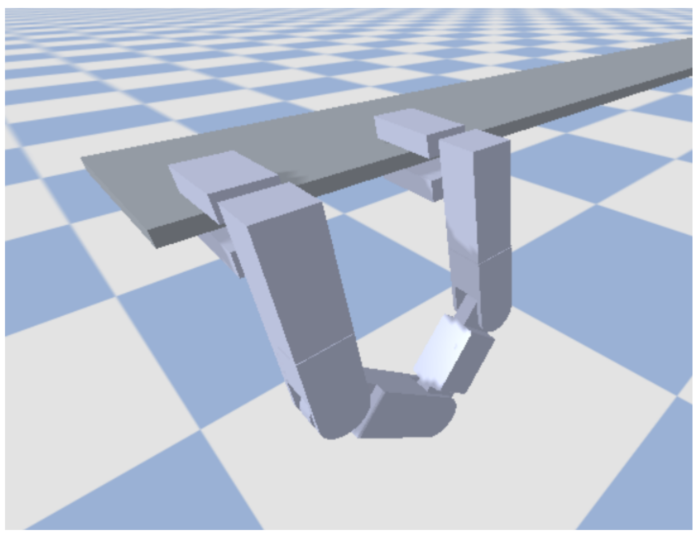
Due by: TBA Prof. Chi-Ying Lin

Ledge-Climbing Robot Control

Objective:

Given a custom **Ledge-Climbing Robot**, implement and compare **DDPG** and **PPO** for ledge-climbing robot control. The goal is to train the robot to accomplish the following four tasks. At least Task 1 and Task 2 must be completed:

1. Complete the horizontal ledge lateral climbing task, where the robot climbs laterally to the right end of the ledge.
2. Complete the horizontal-to-inclined ledge transition lateral climbing task, where the robot climbs laterally to the right end of the ledge.
3. Complete the inclined-to-inclined ledge transition lateral climbing task, where the robot climbs laterally to the right end of the ledge.
4. Select one of Tasks 1–3 and modify it into a reciprocal climbing task, where the robot climbs to the right end first and then returns to the left end.



Common Environment Description:

- Each group must use the assigned design (URDF) for training.
- Each episode starts from the leftmost initial position.
- There are four different ledge configurations in total.
- The state includes joint angles, joint velocities, gripper positions, gripper velocities, middle joint velocity, previous claw actions, and previous wrist actions.
- An initial reference action is provided. The agent learns a residual action to modify the reference trajectory.
- An episode ends when the robot falls from the ledge or completes the path. The maximum number of steps per episode must be clearly reported.
- The reward should encourage successful ledge-climbing behavior. Possible reward terms include forward progress, wrist compensation, gripping posture, and action cost.
- The state, action space, reference action trajectory, reward, or termination conditions may be modified, but any modification must be clearly explained and justified.

Methods

Apply and compare the following methods:

- Deep Deterministic Policy Gradient (DDPG)
- Proximal Policy Optimization (PPO)

Analysis Requirements

You must analyze and discuss the following:

- Compare DDPG and PPO in terms of training stability, convergence speed, final performance, episode length, and success rate.
- Analyze the performance differences among different ledge configurations.
- Discuss whether residual learning improves the reference trajectory.
- Analyze how the learned residual actions affect climbing behavior, gripping posture, and motion stability.
- Explain the reward design and the role of each reward term.
- Show learning curves, including episode return, episode length, success rate, forward progress, losses, and action or residual-action magnitude.
- Demonstrate the learned climbing behavior under different ledge configurations.
- Report the main hyperparameters, including learning rate, discount factor γ, network architecture, batch size, replay buffer size for DDPG, rollout length for PPO, and update frequency.
- Present the assigned configuration, including robot design, joint settings, ledge configuration, action space, observation space, and maximum episode steps.
- If the state, action space, reference action trajectory, reward, or termination conditions are modified, explain and analyze the changes.

Reference

- Andi Alfian Kartika Aji and Chi-Ying Lin, "Efficient learning for active wrist compensation strategy of ledge-climbing robot," Applied Intelligence, vol. 56, article no. 243, 2026. Available: `https://link.springer.com/article/10.1007/s10489-026-07245-7`